Weekly Test

SECONDARY COURSE – MATHS 211

Roll No. **Code No.WT1/XI/211/26**

Set A

Day and Date of Examination ____________________

Signature of Invigilat1ors 1. ____________________

2. ____________________

General Instructions:

1. Candidate must write his/her Roll Number on the first page of the Question Paper.
2. Please check the Question Paper to verify that the total pages and the total number of questions
contained in the Question Paper are the same as those printed on the top of the first page. Also check to see that the questions are in sequential order.
3. For the objective-type of questions, you have to choose any one of the four alternatives given in the question i.e. (A), (B), (C) or (D) and indicate your correct answer in the Answer-Book given to you.
4. All the questions including objective-type questions are to be answered within the allotted time and no separate time limit is fixed for answering objective-type questions.
5. Making any identification mark in the Answer-Book or writing Roll Number anywhere other than the specified places will lead to disqualification of the candidate.
6. In case of any doubt or confusion in the question paper, the English Version will prevail.
7. Write your Question Paper Code No. 70/OS/2, Set A on the Answer-Book.
8. (a) The Question Paper is in English/Hindi medium only. However, if you wish, you can answer in any one of the languages listed below:
English, Hindi, Urdu, Punjabi, Bengali, Tamil, Malayalam, Kannada, Telugu, Marathi, Odia, Gujarati, Konkani, Manipuri, Assamese, Nepali, Kashmiri, Sanskrit and Sindhi.
You are required to indicate the language you have chosen to answer in the box provided in the Answer-Book.

(b) If you choose to write the answer in the language other than Hindi and English, the responsibility for any errors/mistakes in understanding the questions will be yours only.

General Instruction :

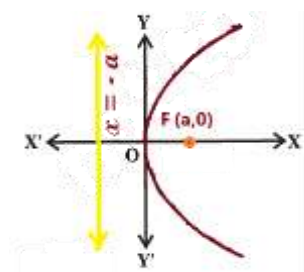
1. Answers of all questions are to be given in the Answer-Book given to you.
2. 15 minutes time has been allotted to read this Question Paper. The question paper will
be distributed at 10.30 a.m. From 10.30 a.m. to 10.45 a.m., the students will read the question paper only and will not write any answer on the Answer-Book during this period.

Note :

(i) This question paper consists of 24 questions in all.
(ii) All questions are compulsory.
(iii) Marks are given against each question.
(iv) Section – A consists of:
(a) Q. No. 1 to 10 – Multiple Choice type Questions (MCQs) carrying 1 mark each. Select and write the most appropriate option out of the four options given in each of these questions.
(b) Q. No. 11 to 15 – Objective type questions. Q. No. 11 to 15 carry 2 marks each (with 2 sub-parts of 1 mark each) and Q. No. 16 carries 5 marks (with 5 sub-parts of 1 mark each). Attempt these questions as per the instructions given for each of the questions 11 to 15.
(v) Section – B consists of:
(a) Q. No. 17 to 21 – Very Short Answer type questions carrying 2 marks each.
(b) Q. No. 22 to 23 – Short Answer type questions carrying 3 marks each.
(c) Q. No. 24 – Long Answer type questions carrying 5 marks each.

Time: 2 Hours **Maximum Marks:** 50

	SECTION A	
	1. If 4x + i(3x − y) = 3 + i (− 6), where x and y are real numbers, then the values of x and y are. (A) $\frac{4}{3}$, $\frac{33}{4}$ (B) $\frac{3}{4}$, $\frac{33}{4}$ (C) $\frac{3}{4}$, $\frac{4}{33}$ (D) $\frac{3}{44}$, $\frac{3}{4}$	1
	2. $(-5i)\left(\frac{1}{8}i\right)$ in the form of a + bi will be (A) $\frac{5}{8}+0i$ (B) $\frac{8}{5}+3i$ (C) $\frac{6}{5}+8i$ (D) $\frac{5}{8}+9i$	1
	3. The slope of lines Making inclination of 60° with the positive direction of x-axis. (A) $-\frac{3}{2}$ (B) 0 (C) $\sqrt{3}$ (D) Not defined	1

	4. The equation of the line through the points (1, −1) and (3, 5) is (A) $x + 3y + 8 = 0$ (B) $-9x + y + 4 = 0$ (C) $\frac{3}{5}x + 5y + 6 = 0$ (D) $-3x + y + 4 = 0$	1
	5. The centre and the radius of the circle $x^2 + y^2 + 8x + 10y - 8 = 0$ (A) centre at (7, −5) and radius -4. (B) centre at (− 4, 7) and radius 5. (C) centre at (−4, −5) and radius 7. (D) centre at (− 7, 5) and radius -4.	1
	6. The equation to the parabola with vertex at the origin, focus at $(a, 0)$,$a > 0$ and directrix $x = -a$ is (A) $y^2 = 4\mathrm{ax}$ (B) $y^3 = 2\mathrm{ax}$ (C) $x^2 = 4\mathrm{a}$ (D) $y^2 = 4\mathrm{a}x^2$ 	1
	7. An ______________ is the set of all points in a plane, the sum of whose distances from two fixed points in the plane is a constant. (A) Circle (B) Ellipse (C) Hyperbola (D) Parabola	1
	8. The point (-3,1,2) lie in ____________ quadrant (A) I (B) II (C) VII (D) VIII	1

9. Which of the following statements represents that the value of $f(x)$ approaches 1 as x approaches 1 from the left? 1

Number line

0 -------- 0.5 ----- 0.9 ----- 0.99 -----> 1

↑

approaching from LEFT

(A) $\lim_{x \to 1^+} f(x)=1$

(B) $\lim_{x \to 1^-} f(x)=1$

(C) $\lim_{x \to 1} f(x)=1$

(D) $f(1) = 1$

10. $f'(x) = \lim_{h \to 0} \frac{f(x+h)-f(x)}{h}$ is called the 1

(A) Product Rule
(B) First Principle of Derivatives
(C) *Definition of the Derivative of a Function*
(D) *Both B and C*

11. Write TRUE for correct statement and FALSE for incorrect statement

(i) The conjugate of the complex number z = a + ib is denoted by z = a - ib 1

(ii) $\lim_{x \to a} \frac{x^n - a^n}{x-a} = na^{n-1}$ 1

(iii) Distance between two points $P(x_1, y_1, z_1)$ & $Q(x_2, y_2, z_2)$ is given by $PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$ 1

	(iv) The eccentricity of a hyperbola is the ratio of the distances from the centre of the hyperbola to one of the foci and to one of the vertices of the hyperbola means $$eccentricity = \frac{vertex}{focus}$$	1
	12. Find the derivative using the principle	
	(i) $cos\left(x - \frac{\pi}{8}\right)$	1
	(ii) $\sin(x+1)$	1
	13. Find the distance between pair of points	
	(i) (2,3,5) and (4,3,1)	1
	(ii) (2,-1,3) and (-2,1,3)	1
	14. Reduce the following equations into slope - intercept form and find their slopes and the y - intercepts.	
	(i) $x + 7y = 0$	1
	(ii) $6x + 3y - 5 = 0$	1
	15. Fill in the blanks	
	(i) The x-axis and y-axis taken together determine a plane known as _______.	1
	(ii) The derivative is the instantaneous rate of change of a function, found by taking the limit of the _______________ quotient as h approaches zero.	1
	(iii) The plane used to represent complex numbers graphically is called the __________ plane.	1
	(iv) The slope of a line making an angle θ with the positive x-axis is ________.	1

16.

Equation	Value of Discriminant (D)
P : $x^2 - 6x + 9 = 0$	0
Q : $x^2 + 5x + 6 = 0$	1
R : $x^2 + 4x + 5 = 0$	−4
S : $2x^2 + 7x + 3 = 0$	25

Based on the above table, answer the following questions.

(i) Which equation has equal and real roots? 1

(A) P

(B) Q

(C) R

(D) S

(ii) Which equation has no real roots? 1

(A) P

(B) Q

(C) R

(D) S

(iii) Which equation has two distinct real roots? 1

(A) P only

(B) Q and S

(C) R only

(D) P and R

(iv) The discriminant of equation S is: 1

(A) 0

(B) −25

(C) 25

(D) 49

	(v) If the discriminant of a quadratic equation is negative, then its roots are: (A) Equal and real (B) Distinct and real (C) Not real (D) One real and one imaginary	1
	17. Find the multiplicative inverse of $\sqrt{5}+3i$	2
	18.In what ratio, the line joining (–1, 1) and (5, 7) is divided by the line x + y = 4?	2
	19. Find the equation of the circle passing through the points (4,1) and (6,5) and whose centre is on the line 4x + y = 16. Or Find the equation of the set of points which are equidistant from the points (1, 2, 3) and (3, 2, –1).	2
	20. Find the derivative of x2 – 2 at x = 10. Or Find the derivative of 99x at x = 100.	2
	21. Find the derivative of $\frac{x^5-\cos x}{\sin x}$ Or Find derivative of $\frac{x+\cos x}{\tan x}$	2

	22. Three vertices of a parallelogram ABCD are A(3,− 1,2), B (1,2, −4) and C (− 1,1,2). Find the coordinates of the fourth vertex.	3
	23. Find the equation of the ellipse, with major axis along the x-axis and passing through the points (4, 3) and (− 1,4). Or Find the modulus of $\frac{1+i}{1-i} - \frac{1-i}{1+i}$	3
	24. A particle moves along a straight line such that its displacement from the origin at time x seconds is given by $s = \sin(x + a)$ where a is a constant. Find the instantaneous velocity of the particle at any time x. Or An engineer models the intensity of vibrations in a rotating machine by the function $I(x) = cosecx\, cotx$ where x is the angular position of the rotating shaft. Determine the rate at which the vibration intensity changes with respect to the angular position.	5